

LECTURE / SESSION¹ PLAN

Name Lecturer: Sofia Gil-Clavel	Date session: 2	Expected number of students: 20
Course title	R tools for teaching and research	
Topic of lecture or session	R-Markdown	
Situational factors (e.g. group size, prior knowledge, expected motivation)	The class lasts 3hrs. The students have at least basic knowledge of R and tidyverse. The students use different Operating Systems (Windows, Mac, or Linux).	
Intended learning outcomes of this session	At the end of the session the students will be able to: 1. Explain what R-Markdown is. 2. Use R-Markdown to generate pdf, Word, and markdown documents. 3. Use the different R-Markdown settings to control the appearance of their documents. 4. Use R-Markdown in class to standardize students' homework outputs.	
Learning material (book, chapters, ...)	The lecture is based on the book: • Xie, Yihui, J. J. Allaire, and Garrett Golemund. "R Markdown: The Definitive Guide." Routledge & CRC Press. Accessed July 21, 2026. https://www.routledge.com/R-Markdown-The-Definitive-Guide/Xie-Allaire-Golemund/p/book/9781138359338. It can be accessed for free here: https://pkg.yihui.org/rmarkdown-book/.	
Media, equipment, tools	The students use their own laptops. The teacher needs access to a projector and a whiteboard.	
Preparation for students	The students have access to the slides and codes before the class: https://github.com/SofiaG11/Course_Rtools/2_Rmarkdown	

¹ A session means a teaching and learning session for university students in the bachelor or master. It can be a lecture, a seminar or a specific type of educational meeting with a group of students that is relevant for the discipline where the lecturer can demonstrate a whole range of teaching skills. It means that in one session each of the six didactic elements appears at least once. Your lesson plan should show that you are using a powerful learning environment and that students are activated.

Time ² (min.)	Didactic element (goal) ³ and topic ⁴	What the teacher does ⁵ (teacher activity)	What students do (learner activity)	Evaluation ⁶ (feedback/assessment)
30	Introduction to R-markdown.	- The teacher uses the slides to explain what R-markdown is and the advantages of using it. - The teacher explains how to install R-Markdown.	- The students passively digest what the teacher is explaining. - The students open R-Studio and follow the instructions in their computers.	- The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student. - The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.
30	Understand and explain the basic logic behind using	The teacher uses the slides to explain the different elements in a R-Markdown file.	- The students passively digest what the teacher is explaining. - The students follow the steps in their computers.	The students will use these concepts during the workshop. So, the teacher will detect when a student confuses

² Indicate the planned duration in minutes

³ State the number(s) of the relevant ILO at each didactic element

⁴ Only key words

⁵ Specify the type of activity and write down also the questions that you prepared to ask in the session

⁶ Specify the type of evaluation, i.e. the way in which you assess if the objective(s) has/have been achieved

	R-Markdown to create pdf and word documents.			them. Based on these confusions the teacher will be able to correct the student. The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.
40	How to change the parameters in the default R-markdown script and how to change them per chunk.	- The teacher shows the different elements using the slides. - The students follow some basic exercises. - The teacher also explains where to look for documentation, how to read it, and how to apply it.	The students follow the steps in their computers.	The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.
40	Use R-Markdown in class to standardize students' homework outputs.	- The teacher explains the logic behind different algorithms but same results. - The teacher leaves two exercises to show how to track students understanding of learning outcomes using R-code.	- The students passively digest what the teacher is explaining. - The students follow the steps in their computers.	- The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student. - The teacher walks around the classroom to check on the

		• The teacher shows how R-Markdown allows tracking learning outcomes.		students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.
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